Sticks and Stones Game Design Doc

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1 Overview

1.1 Vision

Good games express a great truth. Great games express it so well they teach it to the player. Awesome games become many great truths and let the player become them as well.

The truths this game hopes to teach are:

- Human history is in a complex organic relationship with the physical world.

- Simulations are an awesome way to learn things about the world.

The experiential truths this game hopes to become are:

- Complex logistics that can be changed, yet is complex enough that the player experiences the wholeness of reality and the real, open, subjective nature of decision making in it.

- The massive, diverse, a priori diet of the world that feeds the iconic, isolated crystallizations, as moments, legends, philosophies, and art.

- As time progresses, certain new systems (which have their own laws) emerge from the interactions of the previous systems, which creates new forms and qualities of life.

- The scope of view changes everything, and a true life requires zooming in and zooming out.

- Human history is made up of a massive amount of details, yet these coalesce into truly revolutionary moments.

In practice, this game hopes to be:

- A simulation moving in development, technology, and culture from primitive to advanced society, under the player's influence

- A simulation which can take curated data (often in the form of mods) to immerse the player in an actual, specific historical period

- A good balance between realism and the complexity goals laid ahead, without being boring

- A simulation where the NPCs can autonomously produce simulations of the most extreme moments of human life: having and loving tribes and families, making enemies, dying unique deaths, creating art, expressing powerful emotions, having new thoughts and views on life,

connecting to wider groups, but at the same time the player having both a role in and a being able to be sensitive witness to these moments.

- A simulation where complex and interweaving groups that mirror the new world can autonomously arise due to the individual NPCs actions, but at the same time the player having both a role in and a being able to be sensitive witness to the character and actions of these (some of these) groups.

- A simulation where the player goes 'oh wow! I never expected that something that dramatic could happen, but its so believable!'

1.2 Description

'Sticks and Stones' is a 2D Tile Colony Simulator, similar to other indie Colony Simulators like 'Rimworld', 'Dwarf Fortress', and 'Song of Syx.' It is set apart by:

1. Real-world historicity and immersion, utilizing real-world technological and cultural development to give the player a realistic yet open-ended experience through a human history of their own.

2. Complex evolving group dynamics. The player is able to engage with and influence groups which evolve in character and merge and split, from the family, tribe, clan, up to higher level formations like cultures, religions, kingdoms, empires, republics.

3. Complex and evolving technology, logistics, warfare. The detail of the world allows the evolution of culture, alliances, and technology to be seemless and dialectic. What is discovered, what is produced, what is valued by the NPCs, have hard prerequisites yet the order is not set beforehand and the web of technology is so large that every playthrough feels like a unique history.

4. Unity of both the detail of simulation at the individual level (view of one unit, their skills, satisfaction, values, life experience) and the grand world level (view of massive structures, the interest of society, large factions). The larger comes into being as the game progresses.

1.3 Technologies

The engine is coded from scratch using C++, using CMake and clangd, with the help of free software libraries SDL2, cereal, $\text{magic}_{\text{enum}}$, $\text{SDL}_{\text{FontCache}}$, and FastNoiseLite, as well as Jinja2 for Python metaprogramming.

2 World Simulation

2.1 Generation

The world is procedurally generated tile-based game. There is some non-determinism due to the way that altitude is currently determined, which affects later terrain generation. The first stage of the generation happens at the chunk level. First, water bodies and land masses are generated, then mountain ranges and other general altitude zones. Then rivers and lakes. Then humidity is determined by distance from water bodies and randomness. Then Temperature is determined by distance from the equator, water bodies, and randomness. Then biomes based on Temperature, humidity and distance from mountains and water bodies. Then the second stage zooms into the Tile level. Tile altitude is determined based on chunk altitude, biomes, and randomness. Tiles are generated in clusters based on biomes, Temperature, altitude, and humidity. After generation is finished, the chunk and tile data are transferred to permanent databases. Biome data is eliminated as it has no actual effect but is only an intermediate data step used to generate the world.

2.2 Terrain / Tile

There are many different Tiles, where different actions can be performed (building, farming, mining). Each Tile has its own altitude, and liquid content (comes from precipitation, rivers).

2.3 Plant

Grass and Plants grows spontaneously on tiles. Grass growth just changes the Tile type. Some Plants take up more than one tile like large grown trees and grapevines. If plants compete for spaces, one will stop growing. Plant growth depends on the tile having enough water, sunlight, Temperature. Plants can not grow or die due to any of these factors being unoptimal. Plant growth can be boosted by fertilizer. Some Plants produce fruit. Some Plants hibernate and can last for several years.

2.4 Animal

Animals eat Plants and other Animals. Some can be domesticated, some easier then others. Some move in packs, some alone. Many can be used for cookable meat and many produce other useful items from their bodies or corpses. Animals migrate. Some Animals may be Valued by Groups.

2.5 Physics

2.5.1 Temperature

Each chunk has its own base level Temperature value, as well as diurnal (daily) Temperature variance and seasonal Temperature variance, to determine the actual Temperature.

Tiles by default take their Temperature from the chunk, but can have exceptions (usually caused by unit buildings, fires, etc.) The chunk factors also determines precipitation, (it precipitates across entire chunks).

Weather Events can also effect Temperature.

2.5.2 Humidity

Each chunk's base Humidity value is determined by the nearness of water bodies, as well as rain shadow effects from mountains.

Tiles by default take their Temperature from the chunk, but can have exceptions (usually caused by unit buildings, fires, etc.) The chunk factors

also determines precipitation, (it precipitates across entire chunks).

Weather Events can also effect Temperature.

2.5.3 Altitude

Each Tile has an Altitude. Altitude effects Temperature, but is only modified during World generation as well as the landslide, snowfall, flood, and tsunami Weather Events.

2.5.4 Precipitation

The event of Precipitation, as well as its severity, happens based on the Humidity of the chunk, a cycle, and randomness. Precipitation events make Liquid accumulate.

2.5.5 Liquid

Liquid flows from high Altitudes to low Altitudes. Certain thresholds of Liquid create certain water Tiles once they have been reached, returning to the underlying Tile if they have been removed.

2.5.6 Weight

All Items have Weights. Units can only carry so much Weight.

2.5.7 Bulk

All Items have Bulks. Units can only carry so much Bulk, but Containers can increase the amount of absolute Bulk they can carry.

1. Bulk Level The hierarchy of Bulk Level is: liquid, fine, semifine, loose, soft, solid, single. Each Item has a certain Bulk Level, and every Container (which includes the hands of a Unit) can only have one Bulk Level active at once. The 'single' and 'liquid' Bulk Levels are exclusive: a Container can only carry one of those Items at once, and no

other Items. The other levels have limited Bulks depending on the Container in question, especially limited for Unit hands (a Unit can only carry a handful of fine sand, but a bucket can carry much).

Only certain Containers can carry liquid and fine.

2.5.8 Calories

Units and Animals require a certain level of Calories to survive, and below a certain level accumulate debuffs. Plants, Animals, and Food contains calories.

2.5.9 Weather Event

In progress

3 Units and Unit Interaction

3.1 Unit Basics

Units have health, hunger, sex, age, skin datas, Skills, Roles, Inventories, known Places, and subscribed Groups.

3.1.1 Inventory and Equipment

Units have an Underlayer slot for their head, arms, legs, and chest, and an Overlayer slot for these too. They also can several carry Containers, the amount depending on how much Bulk and Weight the Unit can carry, with a max of 8, not counting nested Containers.

3.1.2 Skills

- note: plan to be modified to be more accurate to real life, rather than TTRPGs.

Units have Skills that affect their performance at certain Roles as well as their Values and allegiance to Groups. Skills also have passive effects. Skills can be increased overtime slowly. Certain Roles increase certain Skills extremely slowly. Skills increase from birth to adulthood but decay in old age.

1. Strength Affects some attacks. Affects max carrying Weight.

2. Athleticism Affects movement speed and some attacks.

3. Dexterity Affects some attacks.

4. Intellect Affects rate of leveling.

5. Memory Increases the amount of Role mana available, as well as skill decay in old age.

6. Perception Natural increase to Satisfaction, especially in high Comfort zones. However, more receptive to Fear.

7. Creativity Easier to learn Roles in different categories, and multiple Roles.

8. Charisma Increases opinion faster within Groups that track opinion. More likely to be given Roles that are elected by Groups. Can increase or decrease Fear.

9. Determination Affects rate of leveling and general efficiency (affects the most Roles), as well as makes Units more actively try to achieve their own Goals and Group Goals.

3.1.3 Places

Units and Groups will respect certain Places for certain activities, which are marked by the player or are created automatically based on certain conditions.

3.2 Role

Each Unit can be assigned Roles. Each Role takes a different amount of Role Mana, which a Unit has a set amount of. The efficiency of the tasks being performed by a Unit with a Role depends on the Skills which boost that Role, and the value of those Skills. The player can assign swap Roles, but only after a cooldown, and some Units will not want to be assigned or read-assnged new Roles.

Each Role can perform certain actions, though actions are not visible to the player.

There are a few categories of Roles get Resources, produce Products (take Resources or other Products to produce new Products),

3.2.1 Trainee

A Trainee is a Role that is in progress (though the Unit may already have achieved previous Roles.)

1. Child All units which are born start as Child. Child can assist others in performing tasks but can only add a bonus to the task efficiency and gain some apprentice experience and increase their Skills.

2. Apprentice When a Unit is first assigned a Role it begins as an Apprentice, and each Role has a different Apprentice time. The Unit can perform the task at a much slower rate while they are an Apprentice.

3.2.2 Resource

Certain Items already exist in the world and only involve a Unit attaining them in some manner. They may require other Items to be harvested but they do not require another Item to be consumed to be made. These are handled by Resource Roles, including but not limited to: gathering, fishing, mining, planting, breeding.

Tend to correlate largely to Strength, moderately to Determination, Athleticism, Dexterity, slightly to Perception Skills.

3.2.3 Product

Certain Items require other Items to be consumed to be made, these are Products. Most Items are Products. However, Producers don't just produce Products, they can also produce Buildables from Products and Resources.

Tend to correlate largely to Dexterity, moderately to Memory, Perception, Strength, slightly to Intellect, Creativity, Determination and Athleticism Skills.

3.2.4 Research

Certain Roles do not produce anything tangible (at least primarily, as many Researchers are also Producers), but contribute to society's progression by advancing Knowledge.

Tend to correlate largely to Intellect, Memory, moderately to Determination, Creativity and Charisma (diplomats), slightly to Perception (explorers) Skills.

3.2.5 Art

Certain Roles produce something that is not used primarily to produce more Products or Buildables for survival. Some Art roles produce nothing tangible at all. Regardless, their output is judged more on its Quality then Resource, Product, or Support Roles.

Tend to correlate largely to Creativity, Dexterity, moderately to Memory, Perception, Charisma, Determination, slightly to Intellect Skills.

3.2.6 Support

Certain Roles largely boost things like Order or Trade, or boost the efficiency of other Roles. Some Roles may actually be counter-productive to the overall development or Order, even of Groups they are a part of (such as Bandit, Zealot, Landowner, and Money Lender roles, depending on the situation), and these fall into this class as well.

Tend to correlate moderately to Charisma, Intellect, and Memory Skills.

3.2.7 Training

Certain Roles train other Roles. Oftentimes this will be a secondary category.

Training is boosted directly by the Charisma, Intellect, Memory, and Perception Skills, as well as the trainer having high Skills relevant to the Role, as well as being in the same Groups or having shared Values.

3.2.8 Melee

Tend to correlate highly with Strength, Athleticism, moderately with Dexterity, slightly with Perception, Determination.

3.2.9 Ranged

Tend to correlate highly with Athleticism, Perception, moderately with Dexterity, slightly with Strength, Intellect.

3.2.10 Leader

Boosts entire Groups or Roles. Tend to correlate with highly with Determination, Charisma, slightly with Intellect, Memory, Perception. though also directly correlates to the actual Roles' Skills.

3.3 Motivations

3.3.1 Basic Motivations

1. Hunger
2. Water
3. Comfort
4. Fear

3.3.2 Bonus Motivations

3.3.3 Personality Motivations

1. Moral Perfection
2. Affection
3. Status
4. Uniqueness
5. Expertise
6. Security
7. Excitement
8. Autonomy
9. Peace

3.4 Value/Valuable

A Value is simply a subjective preference for any Valuable. A Valuable is anything that can be conceptualized with reference to the simulation: certain Animals, Items, Plants, Buildables, Motivations, Units with certain traits (physical, Role, Personality), and Groups. Units can be members of Groups and simultaneously negatively Value them. For the sake of complexity and performance, most Values will be unset, and thus the Unit will have the default preference for that Valuable. Above a certain threshold of Value, a Group's Values will come to influence a Unit's Values.

There are some very common Value Patterns. These are abstract formulas for the Values certain Units or Groups are more likely to be generated with or gain over time. For example, Units are more likely to have certain Values if they have certain Roles or Personality traits. Units are also likely to Value the Role they have especially if they a Group which is Values that Role is a Group they Value.

3.5 Interest/Goal

A Goal is tangible task that a Unit or Group tries to achieve. It can sometimes be a single task, such as 'marry X', 'get X Item', 'kill X'. Groups can also have open-ended versions of Goals, called Interests. An Interest is a Goal which is usually never achieved but progress can always be made, such as 'establish trade routes', or 'expand territory.'

3.6 Group

Groups can grow over time and be small or large. They represent things as narrow as families and tribes and as broad as cultures and empires. A Unit likely belongs to a few or even several Groups. Groups may have their own Motivations and Values, and even Goals and Interests. Oftentime a Group can form around something simple like preferences for a collection of Items if the prospective members of that Group value it enough. The makeup and character of Groups also depends heavily on the state of other Groups and their interrelations. As an example some Groups might even form just as an opposition to another Group and immeadiately dissolve if that Group is destroyed (for example, a Group that forms with the sole goal of assassinating

Groups make the game highly dynamic and unpredictable, and give the player.

A Unit's Satisfaction, Goals, Motivations, and Values will often be derivative of the Groups of which they are a member, especially if they Value that Group. Firstly, when given a choice Units are more likely to choose to be a part of Groups that share their Values.

Groups of certain types also have the ability to negotiate with other Groups– for example, families may marry or exchange gifts, tribes and

princedoms may make

3.7 Event

Groups controlled by the player make the player receive Events related to that Group. A player will not receive duplicate Events if two or more of the Groups they control receive the same Event, unless they are opposed in the Event. (this section is to be expanded).

4 Item

As mentioned their are two main classes of Items, Resources and Products. However there are certain traits of Items that serve different functions. The recipe of new Items is unlocked through new Technology.

4.1 Tool

These are Items which are required to make other Items but not consumed in the process of making a single Item (though some deteroriate as they are used).

4.1.1 Weapon

These are Tools which can be used to deal additional damage.

4.2 Art

These Items often do not have a practical production use after they are created but instead give passive bonuses to Units and Groups.

4.3 Containers

These Items carry other Items, often managing Weight and Bulk.

4.4 Wearable

These Items can be equipped and let Units survive at more extreme Temperatures or protect Units from attacks.

4.5 Perishable

These Items can deterioriate without being used at all.

5 Buildables

Walls, Roofs, and Stations which can be worked at, made of different materials. Most cannot be moved, only destroyed, and cannot enter Inventories. They are Products.

6 Order

Determined by dimensions:

1. Legal
2. Administrative
3. Cultural
4. Satisfaction
5. Force

Order applies to Groups larger than a certain size, and the amount of Order required is determined by the Sacrifice required by the Group. If Order is low, it is likely certain Units will become criminals or rebels, killing other units, and it is more likely new Groups will be formed as Units will be

less loyal to current Groups. It is also more likely that those Groups will be attacked. However the player may choose to support a rebellion against or a transformation of the Groups they control.

6.1 Legal

This source of Order is stable and really only hard to keep above a minimum level, or go down quickly, if High Leaders have recently been killed. However, it is hard for it to get to a high level. Raising Legal Order too high also causes Satisfaction Order to decrease, and makes it harder to produce Administrative Order.

- Produced passively by High Leaders
- High Leaders have abilities to raise Legal Order immeadiately, leading to lower Satisfaction
- Produced by Priests, High Priests, Lawyers, Judges, Senators
- Senators can also ensure that Legal Order doesn't increase past a certain amount
- Reduced when new Knowledge is discovered

6.2 Administrative

This source of Order depends on Legal Order. If Legal Order is too low or too high, Administrative Order will also be harder to create. As Legal Order goes up from a certain point, Administrative Order will also decrease, at a smaller rate. High Administrative Order also means that the Force Order requirements are lower. Administrative Order can also be spent to enact certain Leader abilities, and is decently easy to create quickly. Low administrative order will also lower the amount of actual taxes that are collected, despite the set tax rate. Most importantly, Administrative Order affects Production Command.

- High Leaders produce large amounts passively, depending on their personality traits

- Produced by Landowner Subclasses, Bureaucrat, Senator, and Governor

- Raised by ample Royal wealth

- Lowered by large debt

6.3 Cultural

This source of Order is difficult to create rapidly, and depends on many factors. High Cultural Order gives all Units a passive boost, and also unlocks certain Rituals which give further boosts or abilities. The only downside of having Cultural Order super high is that captured foreign units will have a larger debuff before they are assimilated.

- High Leaders produce an amount, depending on their personality traits

- Mostly produced by Art, especially Artifact Artists, which is multiplied based on the varieties, beauty level, and history of that artform in the nation

- Also produced by Shamans, Priests, High Priests

- Modified positively or negatively by the amount of recently discovered Academic Knowledge

- Lowered by low Value adherence, but can get a massive boost if Value adherence becomes too low or there is a rapid change of Values

6.4 Satisfaction

This source of Order is the easiest kind of Order to create rapidly, as long as the player has enough resources and certain types of Units. However, if Satisfaction is quite low it is also likely that the player is lacking in other crucial resources. High Satisfaction also means that the Force Order requirements are lower.

- Depends primarily on meeting Basic Motivations
- Secondarily on Bonus Motivations, Personality Motivations
- Tertiarily on environment beauty
- Lowered by large amounts of soldiers, and somewhat by large amounts of resource workers
- Lowered by high set taxes
- Hugely lowered by mass death, especially of civilians

6.5 Force

This source of Order is also quite difficult to obtain because it is primarily based on army size and unit quality.

- Calculated by the power of army

7 Knowledge

The forms of Knowledge are:

1. Exploration

2. Technology

3. Units

4. Foreign

5. Academic

New Knowledge of all forms will be unlocked passively at a certain rate (assuming certain units are doing other tasks), except Exploration, but the main source of Knowledge is Researchers. Knowledge itself, also affects other stats such as Cultural Order and Legal Order.

with units that can evolve into other units that have not been discovered randomly having a "Eureka" moment rarely, if the required resources to create that new unit are also available to the player.

7.1 Exploration

Explorers discover previously hidden tiles on the map which stay revealed, unlike other units, which will reveal them for a time and then fog of war will reappear. As explorers permanently reveal tiles, Exploration Knowledge increases. Since the world is infinite, Exploration Knowledge is always decreasing at a fixed rate. Maps from foreign nations can also be bought, which increases Exploration Knowledge and reveals areas of the map.

7.2 Academic

This Knowledge functions more like an Order type. Academic Knowledge does nothing by itself, but vastly speeds up the production of other types of Knowledge and Order.

Perhaps the most important function of Academic Knowledge is that it can be used to create certain roles from Students through Trainers and books, and boost unit and technology research. There are also some technologies

which are near impossible to unlock without a certain level of Academic research.

There are four subcategories of Academic Research:

1. Natural
2. Philosophy
3. Mathematics
4. History

The rate at which Academic Knowledge decays is lessened by the production of books.

7.2.1 Natural

AKA "Natural Philosophy", what we would think of as Science.

1. Techs which Require
2. Units which Require
3. Boosts

7.2.2 Philosophy

This boosts Legal and Administrative Order and boosts/allows training for Order Supporters. Also, it is a prerequisite for some Order Tech/Units. If too low, Legal Order will be set at a minimum.

7.2.3 Mathematics

This boosts/allows training for Engineering Tech Researchers, Explorers, Trade, and Structures Producers.

7.2.4 History

This boosts/allows training for Cultural and Legal Order and Foreign Researchers, as well as a small boost to Satisfaction

7.3 Technology

The required tools which units need to do their roles. Sometimes, it will unlock a new role immediately (usually when a new level metal is discovered), but sometimes, the new roles need to be discovered independently of the technology they use (e.g: priest, scribe).

There is a small chance new technology can be invented by related classes, usually by the unit that evolves into the unit that uses or makes that technology, if the required resources to craft that technology are nearby. However, the most reliable way to unlock new technologies is by having a tech researcher near the required materials (which often means a large set of materials), where he will start to "work on" the problem, and then after a randomized amount of time will "discover" the new technology. Alternatively, technology researchers can be given related books instead when they become available, and given enough time will unlock related technologies. Technology can also be given between Groups.

7.4 Role Knowledge

Closely intertwined with technology, but often unlocking a new technology can unlock no Role, one Role, or many Role, and some Roles can be unlocked only ages after the technological requirements have been unlocked. The same discovery processes that apply to tech apply to units, but instead Researchers are assigned to follow around Units which are predecessors to the Role that needs to be unlocked (though of course, unless the player has played the game before, they do not know what the Role is). Like technologies, researchers can also be given books.

7.5 Foreign

This Knowledge allows closer relations with friendlies, or advantages in warfare against enemies. It also boosts Trade.

If a nation is an enemy (at war), only Spies can be sent, but if a nation is a friendly or neutral, all types of Foreign researchers can be sent. If a nation dislikes your nation, the ruler may or may not allow other Foreign Researchers.

If Foreign Research is high, it will also boost Academic, Technology, and Unit Research through Knowledge sharing, or Exploration Research by purchasing maps.

8 Economy

8.1 Production

Tab that analyzes what is produced in a Group and how much. This tab also allows the player to set automatic policies for managing production of that Group if he has certain Roles. These policies automatically lower and raise quotas depending on certain factors. Group members of certain Roles must collect this data for it to be visible, and it can even be inaccurate.

8.2 Trade

Only Groups of a certain size, with certain Interests, and with a certain amount of Roles can get a Trade score. Trade does not actually reflect the absolute state of production in a Group: a Group can have 0 Trade and still produce plenty for its members. Group members of certain Roles must collect this data for it to be visible, and it can even be inaccurate.

8.2.1 Exports

Amalgam of the *amount of exports* and *average profit*.

1. Amount of Exports Affected by:

    - Surplus production (production set aside by player, that is not to be made or used by domestic workers)

- Amount of Trade(Foreign) produced by Trader Units
- Tarriff Rate
- Domestic Opinion of Foreign Nation
- Foreign Opinion of Domestic Nation
- Knowledge (Foreign)
- Trade Deals and Embargos (can set amount to 0)
- Wars or Previous Wars

(a) Tarriff Rate Because tarriffs set by foreign governments are often actually paid by the foreign customers, this affects the amount of exports as well as the profit, since foreign customers are less likely to buy, and domestic traders are les likely to sell.

(b) Trade Deals

(c) Embargos Other Groups can choose to completely limit foreign exports from your nation.

2. Average Profit Affected by:

- Demand of Goods
- Amount of Exports (negatively)
- Amount of Trade(Foreign) produced by Trader Units
- Foreign Opinion of Domestic Nation
- Knowledge (Foreign)

- Tarriff Rate
- Trade Deals

8.2.2 Imports

The amount of useful imports that are purchased by the Group. This number matters less, but the actual products you or your Units are buying likely matters more. However, one the scores that affects it, *average savings*, likely does matter and determines whether the Group will make future imports. Some Leaders can enact Royal Decrees to order quotas of specific imports, or leave it up to Traders based on regular production quotas and Profit Value.

Like Exports, this is an amalgam of *amount of imports* and *average savings*.

1. Amount of Imports The amount of imports the Group is buying. Player can see this in a pie-chart view based on certain general classes of goods. Affected by:
   - Set Quotas by Royal Decrees
   - Foreign Tarriffs placed by player
   - Foreign Opinion
   - Trade(Foreign) produced by Trader units
   - Domestic Opinion of Foreign Nation
   - Trade Deals/Embargos placed by player
   - War/recent war
2. Average Savings Affected by:

- Domestic goods production
- Amount of Imports (positively)
- Foreign Tarriffs placed by player
- Trade(Foreign) produced by Trader units
- Foreign Opinion
- Trade Deals

8.3 Taxes

The amount of taxable production and trade by the Group. Allows player to raise and lower taxes given correct Roles. Relies on Production and Trade data.

9 War/Military

Regular Units can climb over regular walls, but not tall walls. Siege Towers and Siege Ladders allow units to climb over tall walls.

9.1 Player Control

Player can either select units manually, and control them, or can get lieutenants to have 'active recruiting' of certain priorities, which means that units with military roles will automatic join up with their lieutenants

10 Balancing

Balancing Goals

11 Time and Age Goals

These are not hard ages. At most maybe an achievement, does not affect game mechanics at all but is just a tool for dev understanding and communication.

11.1 Paleolithic

~ Goal of how long the age takes in real time at 2x average speed: 5-7 hours

- Starting pawns: Childrearer, Scavenger, Child
  - Scavenger quickly discovers Knapper after finding flint or rocks and being idle
  - First scavengers that gather sticks and wood and flint discover fire, or put stick in lighting fire to make a torch, unlocking Firekeeper
  - Scavenger can become Scout
- Knapper can make primitive spear, discovering Spear Hunter
- When a fire is made, that unlocks Storyteller
- Older pawns become Elder or Shaman
- Elder thinks of Ornamenter from Animals or found bones, Basket Weaver if hemp or flax is found, Woodcutter if have knapped primitive hand axes, Craftsman, or Shelter Builder if have wood and other materials, and Fish Hunter if around Primitive Spearman, exiting the age (at least one original adult must reach maturity in this age)

11.2 Mesolithic

~ Goal of how long the age takes in real time at 2x average speed: 7-12 hours

- Starting pawns: All paleolitic (though a few that do not tree to elder may not be discovered)